ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

DISTANCE LEARNING PROGRAMME

(Academic Session : 2024 - 2025)

JEE (Advanced)
TEST # 01
07-07-2024

JEE(Main + Advanced) : LEADER TEST SERIES / JOINT PACKAGE COURSE

Test Type : Unit Test # 01

ANSWER KEY

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	B	B	D		
SECTION-I (ii)	Q.	5	6	7	8		
	A.	B,D	A,D	A,B,C	A,B,C		
SECTION-I (iii)	Q.	9	10	11	12		
	A.	A	A	B	B		
SECTION-III	Q.	1	2	3	4	5	6
	A.	5	5	2	5	7	3
SECTION-IV	Q.	1	2				
	A.	A->PR,B->QST,C->PRT,D->QS	A->QS,B->R,C->R,D->PT				

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	B	A	D		
SECTION-I (ii)	Q.	5	6	7	8		
	A.	B,D	B,C	A,C,D	B,C		
SECTION-I (iii)	Q.	9	10	11	12		
	A.	A	B	C	C		
SECTION-III	Q.	1	2	3	4	5	6
	A.	8	3	5	3	3	4
SECTION-IV	Q.	1	2				
	A.	A->RS,B->PT,C->PT,D->PQ	A->PRS,B->PQT,C->P,D->PQT				

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	D	A	B		
SECTION-I (ii)	Q.	5	6	7	8		
	A.	A,B	B,C	A,B,C,D	A,B,D		
SECTION-I (iii)	Q.	9	10	11	12		
	A.	A	D	B	A		
SECTION-III	Q.	1	2	3	4	5	6
	A.	5	7	2	2	2	7
SECTION-IV	Q.	1	2				
	A.	A->R,B->P,C->S,D->Q	A->S,B->Q,C->R,D->T				

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HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (C)

$$x = \frac{2\sin 1^o \cos 1^o}{2} = \frac{\sin 2^o}{2}$$

$$= \frac{2 \times \frac{\pi}{180}}{2} = \frac{\pi}{180}$$

2. Ans (B)

$$\frac{F}{A} = y.\frac{\Delta \ell}{\ell}$$

$$[Y] = \frac{F}{A}$$

Now from dimension

$$F = \frac{ML}{T^2}$$

$$L = \frac{F}{M}.T^2$$

$$L^2 = \frac{F^2}{M^2}\left(\frac{V}{A}\right)^4 \quad \therefore T = \frac{V}{A}$$

$$L^2 = \frac{F^2}{M^2A^2}\frac{v^4}{A^2} \qquad F = MA$$

$$L^2 = \frac{V^4}{A^2}$$

$$[Y] = \frac{[F]}{[A]} = F^1 V^{-4} A^2$$

3. Ans (B)

$R^2 = 9p^2 + 4p^2 + 12p^2\cos\theta$ .....(i)

$4R^2 = 36p^2 + 4p^2 + 24p^2\cos\theta$ .....(ii)

from equation (i) and (ii) equating value of R^2

$13p^2 + 12p^2\cos\theta = 10p^2 + 6p^2\cos\theta$

$3p^2 = -6p^2\cos\theta$

$\cos\theta = -\frac{1}{2}$

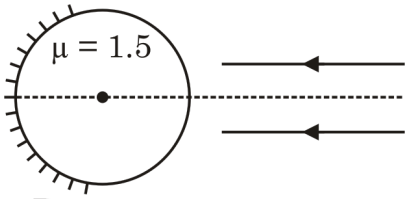
$\theta = 120°$

4. Ans (D)

Refraction : $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

$(u \to \infty)$ $\frac{15}{v} - 0 = \frac{15-1}{40}$

$v = 120$ cm



R = 40cm

Reflection : $v = \frac{uf}{u-f}$

$= \frac{(40)(-20)}{60} = -\frac{40}{3}$

Refraction : $u = 80 - \frac{40}{3}$

$u = \frac{200}{3}$cm

$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

$\frac{1}{v} + \frac{(15)}{(800/3)} = \frac{1-15}{-40}$

$v = -100$ cm

Distance from center:

$= 100 - 40 = 60$ cm (left)

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (B,D)

$\vec{v} = \frac{d\vec{r}}{dt} = 0\hat{i} + 5\left(-\hat{j}\right)$

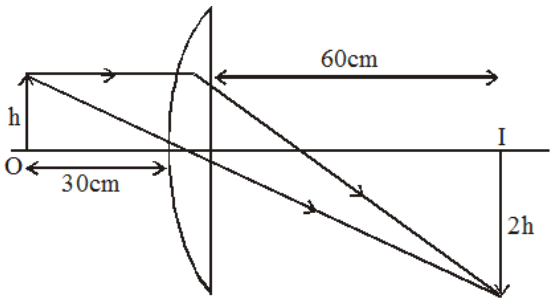
$|\vec{v}| = 5$ m/s

Velocity along $-y$ axis. So parallel to y axis or perpendicular to x axis.

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6. Ans (A,D)

For lens

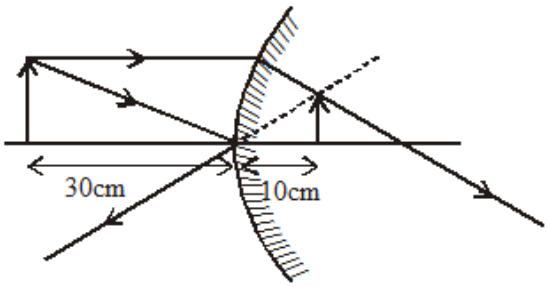


$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\frac{1}{60} - \frac{1}{(-30)} = \frac{1}{f} \Rightarrow f = 20cm$ ... (i)

Also $\frac{1}{f} = (n-1)\left(\frac{1}{R} - \frac{1}{\infty}\right) = \frac{(n-1)}{R}$ ... (ii)

For reflection from convex mirror (curved surface)



$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} = \frac{2}{R}$

$\frac{1}{+10} + \frac{1}{-30} = \frac{1}{f} = \frac{2}{R}$ ... (iii)

R= 30 cm

from (i), (ii) & (iii)

n = 2.5,

faint image erect & virtual

7. Ans (A,B,C)

Given, object is placed at a distance from the pole of a coneave mirror which is equal to the radius of carvature of the mirror.

$u = -R = -2f$

we know, focal length of a concave mirror is negative,

$\frac{1}{f} = \frac{1}{f} + \frac{1}{u}$

we get $\frac{1}{-f} = \frac{1}{v} + \frac{1}{2f}$

or $\frac{1}{v} = \frac{1}{-2f} = -\frac{1}{R}$

$v = -R$

So the image is real, and so inverted and also same sized (as u = v)

8. Ans (A,B,C)

For yellow : $(\mu_g)_y = (\mu_{liq})_y$ = No deviation

For red : $\lambda_R > \lambda_{yellow}$ $(\mu_g)_R > (\mu_{liq})_R$

$\Rightarrow$ Towards base

For blue : $\lambda_b < \lambda_{yellow}$ $(\mu_g)_B < (\mu_{liq})_B$

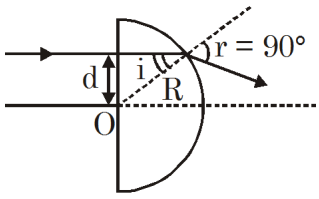
$\Rightarrow$ Towards vertex

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PART-1 : PHYSICS

SECTION-I (iii)

9. Ans (A)



As d increases, 'i' increases,

For d to be maximum, 'i' should be maximum i.e., equal to critical angle

$\Rightarrow < r = 90°$

$\Rightarrow \mu \sin i = 1 \sin 90°$

$\sqrt{2} \sin i = 1$

$\sin i = \frac{1}{\sqrt{2}}$

$\Rightarrow i = 45°$

$\sin i = \frac{d}{R}$

$\Rightarrow d_{max} = \frac{R}{\sqrt{2}}$

10. Ans (A)

For TIR

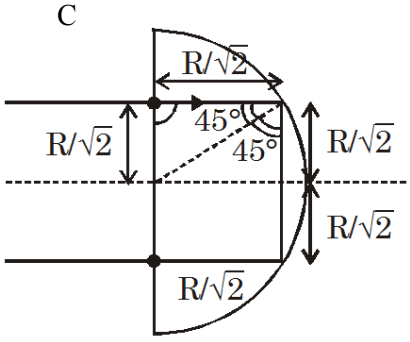
$d > \frac{R}{\sqrt{2}}$

$\Rightarrow d_{min} = \frac{R}{\sqrt{2}}$ (just)

$\mu = \frac{C}{V} = \sqrt{2} \Rightarrow V = \frac{C}{\sqrt{2}}$

$\text{time} = \frac{\text{distance}}{\text{speed}} = \frac{4.\left(\frac{R}{\sqrt{2}}\right)}{\left(\frac{C}{\sqrt{2}}\right)}$

$t = \frac{4R}{C}$



11. Ans (B)

$\vec{A} = 2\hat{i} + 3\hat{j}$ $\quad \vec{B} = -\hat{i} + 2\hat{j}$

$\vec{C} = -3\hat{i} - \hat{j}$ $\quad \vec{D} = +2\hat{i} - 2\hat{j}$

$\vec{A} + \vec{B} + \vec{C} + \vec{D} = 2\hat{j} \uparrow$

12. Ans (B)

$\vec{A} + \vec{C} - \vec{B} = \vec{0}$

PART-1 : PHYSICS

SECTION-III

1. Ans (5)

$2N = 2(\text{kg m s}^{-2})$

$= 2\left(\left(\frac{M}{2}\right)\left(\frac{L}{5}\right)\left(\frac{T}{5}\right)^{-2}\right)$

$= 5MLT^{-2} = 5$ Force

2. Ans (5)

$[\text{Energy}] = ML^2T^{-2}$

In new system

1 unit of energy =

$$\frac{(1 \text{ unit mass})(1 \text{ unit length})^2}{(1 \text{ unit time})^2}$$

$$= \frac{\left(\frac{100}{1000}\text{kg}\right)(4\text{m})^2}{(2\,\text{sec})^2} = \frac{4}{10}\frac{\text{kgm}^2}{\text{s}^2} = \frac{4}{10}\text{J}$$

Now $\frac{4}{10}$J = 1 unit of energy in new system

$1\text{J} = \frac{10}{4}\text{unit}$

$10\text{ J} = \frac{100}{4}\text{ unit} = 25\text{ unit}$

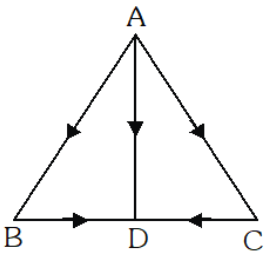
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3. **Ans (2)**

$\vec{AB} + \vec{BD} = \vec{AD}$ ....(1)

$\vec{AC} + \vec{CD} = \vec{AD}$ ....(2)

Δ law



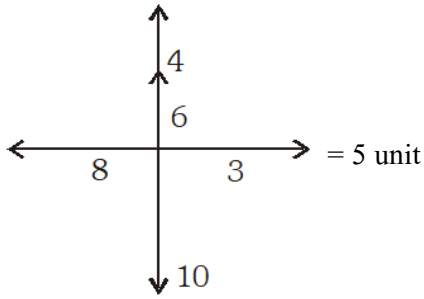
Equation (1) + (2);

$\vec{AB} + \vec{AC} + \vec{BD} + \vec{CD} = 2\vec{AD}$

$\Rightarrow \quad \vec{AB} + \vec{AC} = 2\vec{AD}$

[$\vec{BD} + \vec{CD} = 0$; $\because \vec{BD}$ is equal & opposite to $\vec{CD}$]

4. **Ans (5)**



$\vec{a} = 5\cos 53^\circ + 5\sin 53^\circ$

$\vec{b} = -10\cos 37^\circ + 10\sin 37^\circ$

$\vec{c} = 10\hat{j}$

$\vec{a} + \vec{b} + \vec{c} = -5\hat{i}$

$|\vec{a} + \vec{b} + \vec{c}| = 5$

5. **Ans (7)**

$$\left(\frac{dy}{dx}\right) = \frac{1}{2\sqrt{2x+5}} \times 2 = \frac{1}{\sqrt{2x+5}}$$

at $x = 1$

$$\left(\frac{dy}{dx}\right) = \frac{1}{\sqrt{7}}$$

6. **Ans (3)**

$f_0 = 30$ cm

$L = 27$ cm

$L = f_0 - |f_e|$

a Galilean telescope, the eyepiece lens is concave

Thus, we have

$f_e = f_0 - L = 30 - 27 = 3$ cm

PART-1 : PHYSICS

SECTION-IV

1. **Ans (A->PR,B->QST,C->PRT,D->QS)**

(P) Right Surface ray bends towards normal

$\mu_2 > \mu_1$

left surface undeviated

$\mu_2 = \mu_3$

(Q) Right Surface away from normal

$\mu_1 > \mu_2$

left surface away from normal

$\mu_2 > \mu_3$

(R) Right Surface towards normal

$\mu_2 > \mu_1$

left surface undeviated

$\mu_2 = \mu_3$

(S) Right Surface away from normal

$\mu_1 > \mu_2$

left surface away from normal

$\mu_2 > \mu_3$

(T) Right Surface away from normal

$\mu_1 > \mu_2$

left surface undeviated

$\mu_2 = \mu_3$

(A) $\mu_1 < \mu_2 \longrightarrow$ P, R

(B) $\mu_1 > \mu_2 \longrightarrow$ Q, S,T

(C) $\mu_2 = \mu_3 \longrightarrow$ P, R,T

(D) $\mu_2 > \mu_3 \longrightarrow$ Q, S

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2. Ans (A->QS,B->R,C->R,D->PT)

$\vec{F}_1 + \vec{F}_3 = F$

$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 = \sqrt{3}F$

$\vec{F}_3 + \vec{F}_4 + \vec{F}_5 = 2F$

$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_6 = \sqrt{3}F$

$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 + \vec{F}_5 + \vec{F}_6 = \text{zero}$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (C)

$4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$

$\frac{25.5}{17} = 1.5$ mole 1.5 mole

$2NO + 2CH_4 \rightarrow 2HCN + 2H_2O$

1.5 mole 2 mole 1.5 mole

L.R. = $1.5 \times 27 = 40.5$ g

2. Ans (B)

If the given radius ratio $\left(\frac{r_{A^+}}{r_{B^-}}\right) = (\sqrt{2}-1)$, this implies that cations are present in octahedral voids made by anions. B^- forms fcc lattice and A^+ are present in their octahedral voids.

Now we can say

- Anions are in contact with each other
- Cations are NOT in contact with each other.
- By face diagonal we can say.

$\sqrt{2}\,a = 4r_{B^-}$

$\Rightarrow a = 2\sqrt{2}\,r_{B^-}$

Total volume occupied by all ions

$= 4\left[\frac{4}{3}\pi r_{A^+}^3 + \frac{4}{3}\pi r_{B^+}^3\right]$

$= \frac{6}{3}\pi\left(r_{A^+}^3 + r_{B^-}^3\right) = \frac{16\pi}{3}[5\sqrt{2}-6]$

Packing fraction $= \frac{\frac{16\pi}{3}(5\sqrt{2}-6)r_{B^-}^3}{16\sqrt{2}\,r_{B^-}^3}$

$= \frac{\pi}{3\sqrt{2}}(5\sqrt{2}-6) = 0.79$

PART-2 : CHEMISTRY

SECTION-I (ii)

5. Ans (B,D)

$V \times 20 = 2V \times S \Rightarrow S = 10$vol.

6. Ans (B,C)

Frenkel defect is a dislocation defect which is exhibited by compounds in which cations and anions differs much in their sizes and have low coordination number.

7. Ans (A,C,D)

$4Ag + 8KCN + O_2 + 2H_2O \rightarrow 4K[Ag(CN)_2] + 4KOH$

4×108g Ag $\rightarrow 8 \times 65$g KCN

100g Ag $\rightarrow \frac{8 \times 65}{4 \times 108} \times 100 = 120$g

$\therefore$ 100 Ag dissolves 120g KCN

$\Rightarrow 4 \times 108$g Ag reacts = 32g O_2

1g of Ag reacts = $\frac{32}{4 \times 108} = 0.0740$g

$\therefore$ 100g of Ag reacts = 7.4g of O_2

Volume of O_2 reacts = $\frac{7.4}{32} \times 22.7 = 5.25$ L

8. Ans (B,C)

$P_{total} = X_A P_A^0 + X_B P_B^0$

$P_{total} = 0.5 \times 200 + 0.5 \times 500 = 350$ torr

$P_{obs} < P_{theo}$ (negative deviation)

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PART-2 : CHEMISTRY

SECTION-I (iii)

9. **Ans (A)**

$X_A P_A^o = Y_A P_T$

$$\frac{1}{Y_A} = \frac{1}{P_A^o}\left(P_A^o + \frac{1-X_A}{X_A}P_B^o\right)$$

$$\frac{1}{Y_A} = \left(1 - \frac{P_B^o}{P_A^o}\right) + \frac{P_B^o}{P_A^o}\frac{1}{X_A}$$

10. **Ans (B)**

$$\frac{1}{P} = \left(\frac{1}{P_A^o} - \frac{1}{P_B^o}\right) Y_A + \frac{1}{P_B^o}$$

PART-2 : CHEMISTRY

SECTION-III

3. **Ans (5)**

$$[K^+] = \frac{M_1V_1 \times x + M_2V_2}{x \times V_1 + V_2}$$

$$0.06 = \frac{0.1 \times 5 \times x + 0.02 \times 25}{x \times 5 + 25}$$

$x = 5$

4. **Ans (3)**

CH_2O

$1 + 2 = 3$

5. **Ans (3)**

Given : $\longrightarrow$ Henry law constant, $K_H = 3 \times 10^2$ atm molality $= \frac{5}{9}$ m

means 5 moles of gas in 9 kg solvent (H_2O)

moles of solvent $\frac{9000}{18} = 500$ moles

mole fraction of gas $= \frac{5}{5+500} = \frac{5}{505} = X$

According to Henry law :

Pressure, $P = K_H X$

$$P = 3 \times 10^2 \times \frac{5}{505} = 2.97 \approx \boxed{3}$$

6. **Ans (4)**

$$2HCl + CaCO_3 \longrightarrow CaCl_2 + CO_2 + H_2O$$

2.5 gm $\qquad$ $n = \frac{1.1}{44} = 0.025$

$$\text{Mass} = \frac{2.5}{62.5} \times 100 = 4\%$$

PART-2 : CHEMISTRY

SECTION-IV

2. **Ans (A->PRS,B->PQT,C->P,D->PQT)**

(A) Acetone + $CHCl_3$

–ve devation from Raoult's law $\Delta S > 0$

$\Delta H < 0 \quad \Delta V < 0$

Maximum Boiling Azeotropes.

(B) Ethanol + Water

+ve Deviation from Raoult's law $\Delta S > 0$

$\Delta H > 0 \quad \Delta V > 0$

Minimum Boiling Azeotropes

(C) $C_2H_5Br + C_2H_5I$

Ideal solution No Azeotropes

$\Delta H = 0 \quad \Delta V = 0$

(D) Acetone + Benzene

+ve Deviation from Raoults law

$\Delta H > 0 \quad \Delta V > 0$

$\Delta S > 0$

Minimum Boiling Azeotropes

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (C)**

$\sin^{-1}(1-x) - 3\sin^{-1}x = \cos^{-1}x$

$\sin^{-1}(1-x) - 3\sin^{-1}x = \frac{\pi}{2} - \sin^{-1}x$

$\sin^{-1}(1-x) = \frac{\pi}{2} + 2\sin^{-1}x$

$2\sin^{-1}x = -\cos^{-1}(1-x)$ .....(1)

Domain is $x \in [0, 1]$

$\sin^{-1}x = \theta,\ \theta \in \left[0, \frac{\pi}{2}\right]$

$x = \sin\theta$ .....(2)

By (1) and (2)

$2\theta = -\cos^{-1}(1 - \sin\theta)$

$\cos^{-1}(1-\sin\theta) = -2\theta,\ -2\theta \in [-\lambda, 0]$

Only possible when $-2\theta = 0$.

$\Rightarrow \theta = 0 \quad \Rightarrow x = 0.$

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2. Ans (D)

$$\sin^{-1}(2x) + \frac{\pi}{6} \geq 0$$

$$\Rightarrow \sin^{-1}(2x) \geq \frac{-\pi}{6}$$

$$\Rightarrow 2x \geq \frac{-1}{2}$$

$$\therefore \frac{-1}{4} \leqslant x \leqslant \frac{1}{2}$$

3. Ans (A)

$$fof(x) = \frac{f(x)-3}{f(x)+1} = \frac{\frac{x-3}{x+1}-3}{\frac{x-3}{x+1}+1}$$

$$fof(x) = -\frac{x+3}{x-1}$$

$$fofof(x) = \frac{-\frac{x+3}{x-1}-3}{-\frac{x+3}{x-1}+1} = x$$

4. Ans (B)

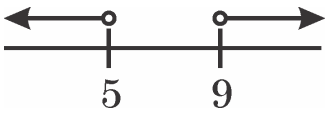
$$y = \frac{x^2+34x-71}{x^2+2x-7}$$

$$\Rightarrow (y-1)x^2 + (2y-34)x - (7y-71) = 0$$

for $x \in \mathbb{R}$

$$(y-17)^2 + (y-1)(7y-71) \geq 0$$

$$y^2 + 289 - 34y + 7y^2 - 78y + 71 \geq 0$$



$$8y^2 - 112y + 360 \geq 0$$

$$y^2 - 14y + 45 \geq 0$$

$$(y-5)(y-9) \geq 0$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,B)

$$S_n(x) = \sum_{k=1}^{n} \tan^{-1}\left(\frac{x}{1+kx(kx+x)}\right)$$

$$= \sum_{k=1}^{n} \tan^{-1}\left(\frac{(kx+x)-(kx)}{1+(kx+x)(kx)}\right)$$

$$S_n(x) = \tan^{-1}(nx+x) - \tan^{-1}x$$

$$= \tan^{-1}\left(\frac{nx}{1+(n+1)x^2}\right)$$

(A) $S_{10}(x) = \tan^{-1}$

$$\frac{10x}{1+11x^2} = \frac{\pi}{2} - \tan^{-1}\left(\frac{1+11x^2}{10x}\right) (x>0)$$

(B) $\lim_{n\to\infty} \cot(S_n(x)) =$

$$\lim_{n\to\infty} \frac{\frac{1}{n} + \left(1+\frac{1}{n}\right)x^2}{x} = x\,(x>0)$$

(C) $S_3(x) = \tan^{-1}\frac{3x}{1+4x^2} = \frac{\pi}{4}$

$\Rightarrow 4x^2 - 3x + 1 = 0 \Rightarrow x \notin \mathbb{R}$

(D) $\tan(S_n(x)) = \frac{nx}{1+(n+1)x^2}$; $\forall\, n \geq 1\,;\, x>0$

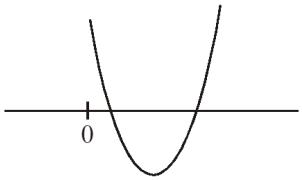
We need to check the validity of

$$\frac{nx}{1+(n+1)x^2} \leqslant \frac{1}{2} \text{ " } n \geq 1\,;\, x>0\,;\, n \in \mathbb{N}$$

$\Rightarrow 2nx \leq (n+1)x^2 + 1$

$\Rightarrow (n+1)x^2 - 2nx + 1 \geq 0\ \forall\ n \geq 1\,;\, x>0\,;\, n \in \mathbb{N}$

Discriminant of $y = (n+1)x^2 - 2nx + 1$ is



$D = 4n^2 - 4(n+1)$ and $n \in \mathbb{N}$

$D < 0$ for $n = 1$; true for $x > 0$

$D > 0$ for $n \geq 2 \Rightarrow \exists$ some $x > 0$

for which $y < 0$ as both roots of

$y = 0$ will be positive.

$y = (n+1)x^2 - 2nx + 1,\ n \geq 2$

So, $y \geq 0\ \forall\ n \geq 1\ ;\ \forall\ x > 0\ ;\ n \in N$ is false.

6. Ans (B,C)

From $D > 0\ \ \forall\ b \in R$.

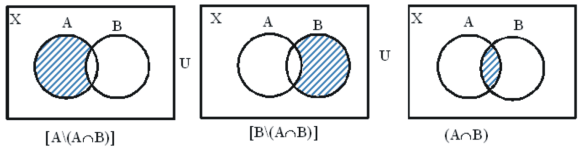
$$a < \frac{5}{8}$$

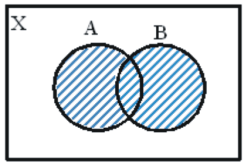
Now verify alternatives.

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7. **Ans (A,B,C,D)**

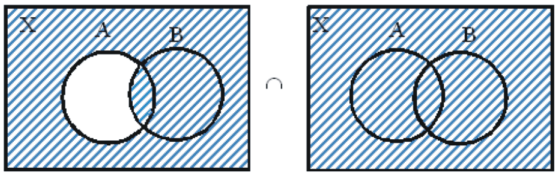
(A) Correct

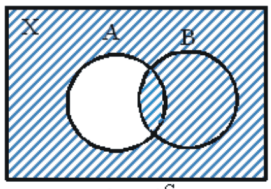




(A∪B)

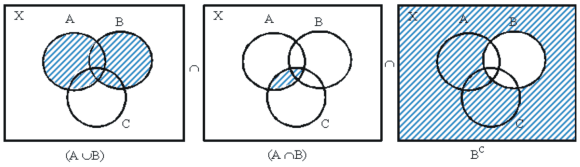
(B) Correct

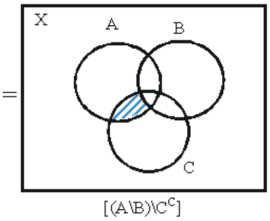




$(A-B)^C$

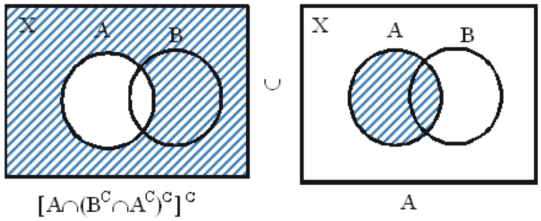
(C) Correct

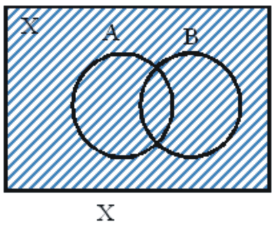




$[(A\setminus B)\setminus C^C]$

(D) Correct





X

8. **Ans (A,B,D)**

(A) Clearly $7^{1/7}, (42)^{1/4} > 1$

$7^{1/7} = 49^{1/14} > 42^{1/14}$

$\therefore 7^{1/7} > (42)^{1/14} > 1$

(B) $\log_3 5 > 0, \log_7 9 > 0, \log_{11} 13 > 0$

$\Rightarrow \log_3 5.\log_7 9.\log_{11} 13 > -2$

(C) $99 + 70\sqrt{2} = 7^2 + \left(5\sqrt{2}\right)^2 + 2.5\sqrt{2}.7$

$= \left(7 + 5\sqrt{2}\right)^2$

$99 - 70\sqrt{2} = \left(7 - 5\sqrt{2}\right)^2$

$\sqrt{99 + 70\sqrt{2}} = 7 + 5\sqrt{2}\ \ 5\sqrt{2} > 7$

$\sqrt{99 - 70\sqrt{2}} = \left|7 - 5\sqrt{2}\right| = 5\sqrt{2} - 7$

$\Rightarrow \sqrt{99 + 70\sqrt{2}} + \sqrt{99 - 70\sqrt{2}} = 10\sqrt{2}$

(D) $\frac{1}{\log_4 3} + \frac{1}{\log_7 3}\ \frac{1}{\log_b a} = \log_a b$

$= \log_3 4 + \log_3 7 = \log_3 28 > \log_3 27 > 3$

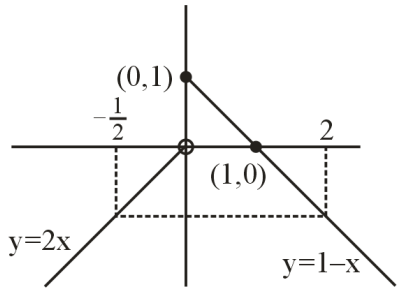
$\therefore$ A,B,D are correct.

PART-3 : MATHEMATICS

SECTION-I (iii)

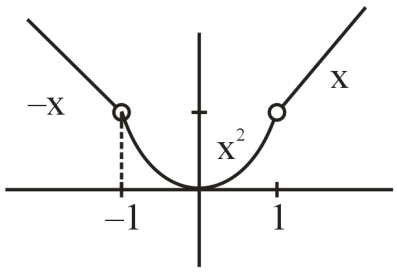
9. **Ans (A)**

f(x)

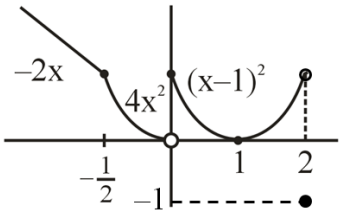


Range $\in (-\infty, 1]$

g(x)



g(f(x))

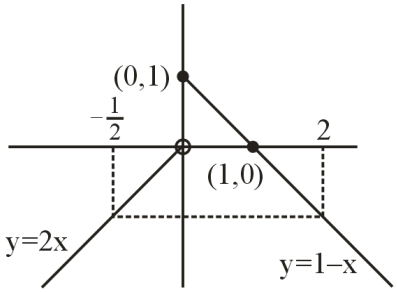


Range of g(f(x)) is $[0, \infty)$

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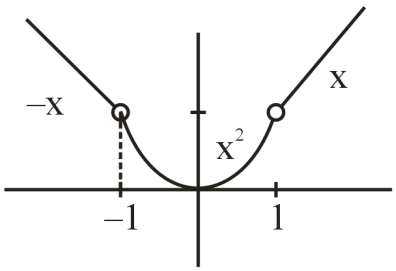
10. **Ans (D)**

f(x)

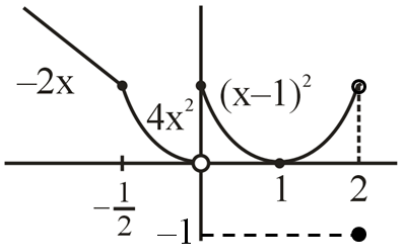


Range $\in (-\infty, 1]$

g(x)



g(*f*(x))



Range of g(*f*(x)) is $[0, \infty)$

11. **Ans (B)**

Let $x = \tan\theta$ or $\theta = \tan^{-1}x$

$$\tan(2\tan^{-1}x) = \tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta} = \frac{2x}{1-x^2}$$

Let $\sin^{-1}x = \alpha$

$\Rightarrow \sin(2\sin^{-1}x) = \sin 2\alpha = 2\sin\alpha\cos\alpha$

$= 2x\sqrt{1-x^2}$

$$\Rightarrow f(X) = \frac{4x^2}{\sqrt{1-x^2}}$$

$$f'(x) = \frac{8x\sqrt{1-x^2} - \left(\frac{4x^2(-2x)}{2\sqrt{1-x^2}}\right)}{\left(\sqrt{1-x^2}\right)^2}$$

$\Rightarrow f'(0) = 0$

12. **Ans (A)**

Let $x = \tan q$ or $\theta = \tan^{-1}x$

$$\tan(2\tan^{-1}x) = \tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta} = \frac{2x}{1-x^2}$$

Let $\sin^{-1}x = \alpha$

$\Rightarrow \sin(2\sin^{-1}x) = \sin^2\alpha = 2\sin\alpha\cos\alpha$

$= 2x\sqrt{1-x^2}$

$$\Rightarrow f(X) = \frac{4x^2}{\sqrt{1-x^2}}$$

$$\frac{4x^2}{\sqrt{1-x^2}} = \frac{16}{\sqrt{1-x^2}}$$

$\Rightarrow x = \pm 2$ (but not in the domain)

PART-3 : MATHEMATICS

SECTION-III

1. **Ans (5)**

$\alpha + \beta = 1,\ \alpha\beta = 2$

$\alpha^2 + \beta^2 = -3$

$\alpha^3 + \beta^3 = -5$

$\alpha^4 + \beta^4 = 1$

$\alpha^5 + \beta^5 = 11$

$$\frac{\sum_{n=1}^{5} S_n}{\sum_{n=1}^{4}\alpha^n + \sum_{n=1}^{4}\beta^n + 7}$$

$$= \frac{S_1 + S_2 + S_3 + S_4 + S_5}{S_1 + S_2 + S_3 + S_4 + 7} = 5$$

2. **Ans (7)**

We have $(5-\alpha)x^2 - 4x + (5-\alpha) \le 0\ ;\ \forall\ x \in R$

$\therefore 5 - \alpha < 0 \quad \text{Disc} \le 0$

$\alpha > 5 \quad \& \ 16 < 4(\alpha - 5)^2$

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3. **Ans (2)**

Bengal white Tigers = BWT

Bengal Royal Tigers = BRT

Males = M

Females = F

Given :

n(BWT) = 6, n(BRT) = 6

n(M) = 5, n(BRT∪M) = 10

n(BRT∪M) = n(BRT) + n(M) − n(BRT∩M)

⇒ 10 = 6 + 5 − n (BRT∩M)

⇒ n(BRT∩M) = 1

⇒ n(BRT∩F) = n(BRT) − n(BRT∩M) = 5

Total Tigers = 12, Male Tigers = 5, Female

Tigers = 12 − 5 = 7

Female Tigers = n(BRT∩F) + n(BWT ∩F)

⇒ 7 = 5 + n (BWT∩F)

⇒n(BWT∩F) = 7 − 5 = 2

Number of Female Bengal Tigers in the Zoo = 2

4. **Ans (2)**

$a^2 + 9b^2 + 6a + 18b + 18 \le 0$

$\Rightarrow \quad (a^2 + 6a + 9) + 9b^2 + 18b + 9 \le 0$

$\Rightarrow \quad (a + 3)^2 + 9(b^2 + 2b + 1) \le 0$

Only possible when $a = -3, b = -1$

then $b - a = -1 + 3 = 2$

5. **Ans (2)**

$$2f(x) + f\left(\frac{x+1}{x-1}\right) = x$$

$$x \to \frac{x+1}{x-1}$$

$$\Rightarrow f(x) + 2f\left(\frac{x+1}{x-1}\right) = \frac{x+1}{x-1}$$

Solving

$$3f(x) = 2x - \frac{x+1}{x-1}$$

$$3f(x) + \frac{x+1}{x-1} = 2x$$

$$\Rightarrow \lambda = 2$$

6. **Ans (7)**

$$a = \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right) + \cos^{-1}\left(-\frac{1}{2}\right)$$

$$= -\frac{\pi}{4} + \pi - \frac{\pi}{3} = \frac{5\pi}{12}$$

$$b = \tan^{-1}(-\sqrt{3}) - \cot^{-1}\left(-\frac{1}{\sqrt{3}}\right)$$

$$= -\frac{\pi}{3} - \left(\pi - \cot^{-1}\left(\frac{1}{\sqrt{3}}\right)\right)$$

$$= -\pi$$

$$a + b = \frac{5\pi}{12} - \pi = -\frac{7\pi}{12}$$

PART-3 : MATHEMATICS

SECTION-IV

1. **Ans (A->R,B->P,C->S,D->Q)**

Do yourself

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2. Ans (A->S,B->Q,C->R,D->T)

$$\left.\begin{matrix}\frac{px^2}{q}+\frac{qx}{p}+p=0\end{matrix}\right\}\begin{matrix}\alpha\\ \beta\end{matrix}$$

$$\alpha+\beta=-\frac{q^2}{p^2},\ \alpha\beta=q$$

(A) $\frac{1}{\alpha}+\frac{1}{\beta}=\frac{\left(\frac{-q^2}{p^2}\right)}{(q)}=-\frac{q}{p^2}$

(B) $\frac{1}{\alpha^2}+\frac{1}{\beta^2}=\frac{\left(-\frac{q^2}{p^2}\right)^2-2q}{q^2}=\frac{q^3-2p^4}{qp^4}$

(C) $|\alpha-\beta|=\sqrt{\frac{\frac{q^2}{p^2}-\frac{4p^2}{q}}{\frac{p^2}{q^2}}}=\frac{\sqrt{q^4-4p^4q}}{p^2}$

(D) $\alpha^3+\beta^3=\left(-\frac{q^2}{p^2}\right)^3-3\left(-\frac{q^2}{p^2}\right)(q)$

$=\frac{q^3}{p^2}\left(3-\frac{q^3}{p^4}\right)$

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